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Seda Özçetin
Umeå Institute of Design, Umeå University, Sweden

Johan Redström
Umeå Institute of Design, Umeå University, Sweden

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Rethinking 'Terms of Service' through programmatic time travel

Seda Özçetin*, Johan Redström

Umeå Institute of Design, Umeå University, Sweden

*Corresponding e-mail: seda.ozcetin@umu.se

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Abstract: The contexts of design are constantly changing, sometimes to the extent that once a 'good' designerly response to an issue, over time becomes increasingly problematic. Therefore, there is often a need to rethink design and its concepts. Programmatic design research may provide an exploratory space for inquiry through specific examples in relation to certain theoretical and conceptual framings. In this paper, we explore [dis/re]orientations towards design histories for creating alternative programmatic research spaces. We work through an everyday challenge, 'Terms of Service' (ToS), a regulatory mechanism amplifying power asymmetries in relating to data-intensive things. Disorienting design by making an odd association between today's ToS and the 'ornament' in early industrial design, we explore resulting reorientations to rethink designing in this domain. Finally, we outline how [dis/re]orientations could be considered a speculative method for making a kind of 'Programmatic Time Travel', using reflections of pasts to reimagine designing for just futures.

Keywords: terms of service; [dis/re]orientations; programmatic research; design histories; just futures

1. Introduction

One of the most important lessons learned when design first turned to larger systems, is that how a system appears is completely dependent on from where you perceive it, and that it therefore is nearly impossible to get a 'complete' view of what it is. To design in the context of complex systems is therefore highly dependent on what perspectives we, consciously or unconsciously, choose to work from. Over time, these choices stabilize, and we might even forget that there are also other ones, that there are many potential entry points into such design that we perhaps did not even consider were there. In this paper, we will explore one such, at first perhaps seemingly odd, entry point into designing for living with computational systems operating at a planetary scale.

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The combination of computation, automation, networking, sensing and actuation, and now machine learning in data-driven processes and platforms have fundamentally changed what 'things' are and capable of (Hauser et al, 2021). In many ways, these new 'things' we refer to as apps, social media platforms, etc. represent a shift from stable objects to a kind of 'fluid assemblages', as "they are made out of a diverse range of material and immaterial resources both contained within the object as it appears in front of us as well as located elsewhere in the network; fluid because their precise forms are assembled dynamically and thus change continuously." (Redström & Wiltse, 2019, p.18). Clearly, such *fluid assemblages* introduce new ways of using, living with, and relating to things – and yet, many of the relations we design for are still framed by concepts with roots in a different time (Hauser et al, 2021).

In what follows, we will explore a different entry point for design into these assemblages. Taking our starting point in the notion of relations rather than functions, we looked for places where these relations are 'designed' but that design seems to have more or less been 'forgotten' about. One such place is the Terms of Service (ToS) that now governs the ongoing relations with things. It has even replaced one of the most central relations we have had with things: ownership. Indeed, to capture how extensive this regulation of relationships has become, it is not even enough to consider the ToS of just one service, since they are increasingly nested and intertwined as a reflection of the data exchange taking place across networks and platforms. To remind ourselves about the difference between a traditional and fixed legal contract and these new dynamic and ever-growing ecosystems of policies and legal frameworks, we propose the term *ToSspheres* (Özçetin, 2023; Özçetin & Wiltse, 2023), suggesting that this is something we live inside.

As an initial indicator that something might be seriously odd here, consider the following: How come that many of the most important aspects regulating our relationship with technologies we do not just use but practically cannot live without, on one side has the expression of endless legal text, and on the other, a simple single checkbox stating, "I agree"? More disturbingly, how come design's primary response has been to streamline these touchpoints rather than questioning this mechanism itself and how it comes to expression?

If the purpose of *ToSspheres* is to perform transparency in data practices so that users comprehend and make decisions to achieve democratic data governance, then, it is necessary that we reimagine how we relate to these things and how we form, sustain, and break these relations. The purpose of this paper is to explore how we can move away from the limited problem framing aiming for frictionless interactions that now seems to guide design. As an antidote to something that is stuck, we will try moving and see if the notion of *[dis/re]orientations* could help us in creating alternative programmatic research spaces.

This inquiry will begin with an introduction to the idea of disorientating design towards design histories. We will then take a close look at ToS and suggest disorientations for arriving at a reorientation. This will be followed by our reflections that draw out tactics of a speculative method for making a kind of *Programmatic Time Travel*. We will continue with a

discussion on contributions and end with concluding remarks inviting further *Programmatic Time Travels*.

2. Disorientating design

Setting a brief for the design inquiry, we started with the following proposition: the changing contexts with *fluid assemblages* require rethinking design and its concepts as well its practices to find alternative orientations. To do this, we take inspiration from Sara Ahmed, who argues that in order to be orientated one might need to first disorientate:

> "When we are orientated, we might not even notice that we are orientated: we might not even think ''to think'' about this point. When we experience disorientation, we might notice orientation as something we do not have.… It is in this mode of disorientation that one might begin to wonder: What does it mean to be orientated?" (Ahmed, 2006, pp.5-6)

We take this question of orientation in relation to temporality, since design has a future orientation as Clive Dilnot (2015, p.165) states "After 1900 to design is to design for the future, it is to bring the future into being as a contemporary possibility.". Observing that this future orientation is rather taken for granted, we act as some sort of 'design therapists' and propose reflecting over pasts to find alternative future orientations for designing with fluid assemblages.

This turn toward pasts is exploratory and reflective, as we do not know from the beginning, where exactly we are turning to or what we will find out. Indeed, as Ahmed (2006, pp.16-17) describes it: "Depending on which way one turns, different worlds might even come into view." In the context of design, this resonates with how design research programs have been used to explore (and distort) different directions for *designing*. Considering how design can be said to engage across a spectrum between particular to general – or from what a *design* is to what *designing* is – we can turn to 'program', where a set of beliefs is tested and reflected upon through specific examples proposing what *designing* could be like (Redström, 2017) (Figure 1). However, instead of aiming for yet another future orientation, we might instead seek to disorientate ourselves by looking at past programs to look for tensions between how certain beliefs were first explored in contrast to how they came to be stabilized over time as they became a "protective belt surrounding the hard core" of programs (Redström, 2017, p.87). These kinds of reflections might then reveal alternative trajectories for defining new programs.

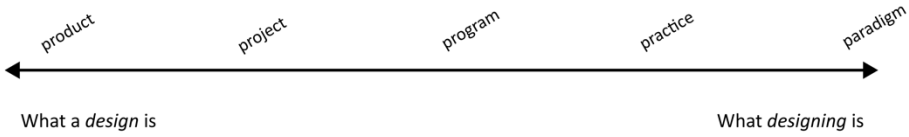


Figure 1 Product – Paradigm spectrum (Redström 2017, p.39).

3. The thing about Terms of Service

In the introduction, we proposed that the existing notion of ToS might have to be replaced in the context of design with the notion *ToSspheres,* highly dynamic and ever-growing ecosystems of policies and legal frameworks, something we live inside, that we repeatedly need to approve and submit, to be allowed to stay and continue to use the products (Özçetin, 2023; Özçetin & Wiltse, 2023) (Figure 2). *Relational mappings,* created through careful readings of ToS, Privacy Policy, and Cookies Policy of services, reveal the actors making these ecosystems and the relations among them (Özçetin & Wiltse, 2023). These overwhelming ecosystems and the data practices that take place within, however, remain invisible as a result of the frictionless interfaces that make the onboarding to services instantly possible (Figure 2). Thus, *ToSspheres,* despite being the main mechanism for governing our relations with fluid assemblages, are known to fail this task when looked from the perspective of a meaningful interaction as they are mostly ignored with a quick check on 'I agree' boxes (Bakos et al., 2014; Maronick, 2014; Obar & Oeldorf-Hirsch, 2018).

Criticisms of *ToSspheres'* long, complex language with technical and legal details resulting in low user engagement have led to explorations into improving ToS' accessibility, readability, and comprehensibility (Bakos et al., 2021; Taber et al., 2020; Reinhardt et al., 2021; Kelley et al., 2009; Kelley et al., 2010; Luger et al., 2013) through the use of visual storytelling, information hierarchy, modularity, typography, and simplification of textual content such as the following: Textured agreements employing visual design techniques (Kay and Terry, 2010), icons for privacy notice and choices (Habib et al., 2021), comics to increase user attention (Tabassum et al., 2018), privacy policy nutrition labels (Kelley et al., 2009; Kelley et al., 2010), visual interactive privacy policies (Reinhardt et al., 2021), modular privacy policies offering more control (Das et al., 2018), data packages introduced through a visual interface (Jones et al., 2017), a crowdsourced sentiment-enhanced visualization of ToS highlighting positive and negative parts (Taber et al., 2020), simplified text stripped from its heavy legalese for better comprehension (Robinson and Zhu, 2020).

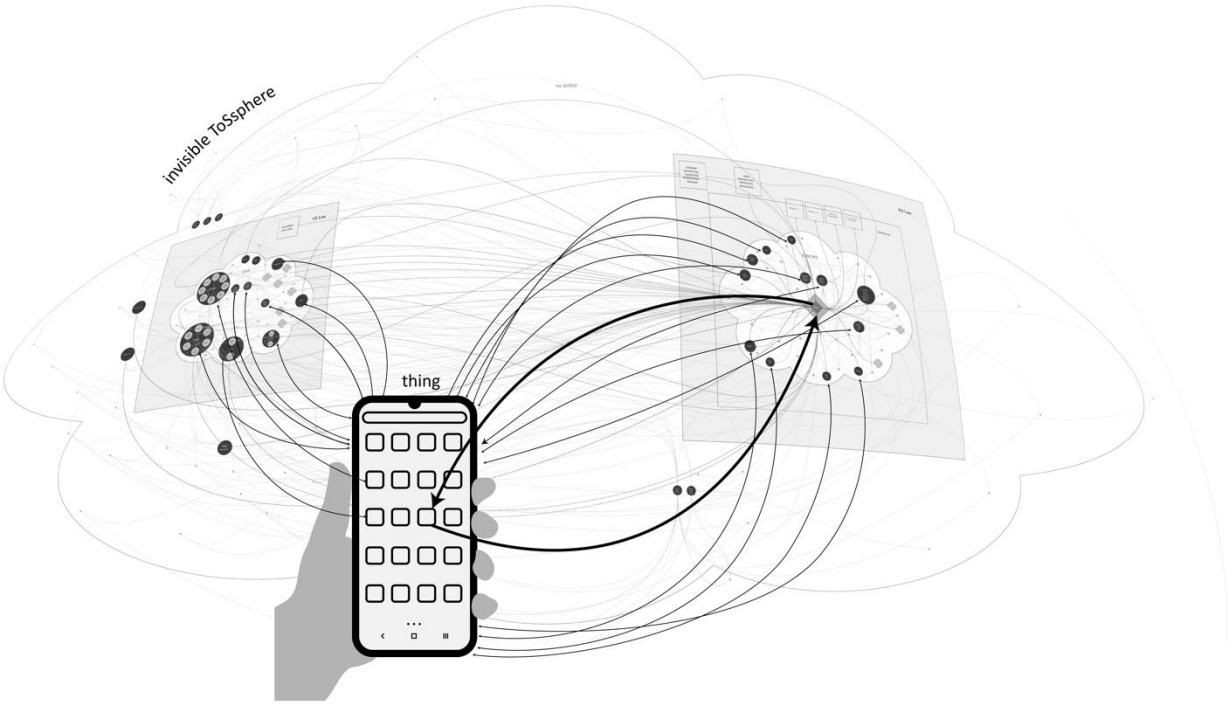


Figure 2 Complexity of ToSspheres

However, these seemingly good explorations report limited gains in engagement measured in few seconds as well as in comprehension. If anything, these experiments, failing to question and understand ToS' complexity, what it is and does, and its relevance, prove that there is a need for different types of inquiries that could open up programmatic research spaces that explore alternative worldviews.

3.1 Disorientating ToS towards Pasts

In constructing an alternative programmatic research space, we look at the experience of *ToSspheres* in use. Several studies mentioning low user engagement with *ToSsphere*s report that users often argue that they don't see the value in reading long and complex *ToSspheres* (Luger et al., 2013; Reinhardt et al. 2021; Tabassum et al., 2018; Taber et al, 2020). In fact, an old study from 2008 calculated the average time an average American spends per year on reading Privacy Policies of visited websites as 244 hours (McDonald & Cranor, 2008). Such study conducted today would find the hours more due to the much more complex *ToSsphere*s. So, could we say that *ToSsphere*s are experienced as *cluttering* and *redundant*?

Moreover, while it is not possible to engage with *ToSsphere*s for their length and complexity, let alone to negotiate their terms, they are presented to people as if a choice and an exercise of agency. Diver (2020, p.2) states that "a real and primary affordance of text is the delay between representation, understanding, and action, a space that might be referred to as a hermeneutic gap". He claims through this delay we have space to contemplate, comprehend, and interpret. Thus, there is value in engaging with a legal text at its own

rather slow pace. This might then suggest that *ToSspheres*, pretending to facilitate informed consent despite not being fit for a fast-paced interaction, are experienced as *dishonest* emphasizing its *redundancy*.

Furthermore, unlike any other part of digital platforms, *ToSsphere*s, with little to no user experience design effort given, do not seem to represent their respective brand identities. This disconnection from the rest of their brands might then make them be perceived as an *afterthought* in the design and production process.

The experience of *ToSspheres* in use as *cluttering*, *redundant*, *dishonest*, and *afterthought* reminds us of the concept of ornament in early industrial design, which has become a topic of discussion and exploration as a result of the shift from craft-based production to industrial. While ornament in craft had value because it represented mastery and labor and it made sense for the techniques, ornament in early industrialization neither had the same value of making nor the relevance to the techniques resulting in its critique for its dishonesty (appearing as if a craft object), its use value (cluttering and redundancy), its irrelevance to the industrial production methods (being an afterthought) (Heskett, 1980). Victoria and Albert Museum's (2024) collection from the 18th century onwards offers many examples of different degrees of the use of ornament in early industrialization.

If we see *ToSspheres* as ornaments of digital interfaces, then it might be insightful to disorientate towards a point in design histories that worked through this kind of a problem space. With respect to 'ornament' and how it was contested in early Modernism when industrial design first emerged, MoMA's 1934 Machine Art exhibition (MoMA, 2016) stands out as a key contribution to design discourse with its exploration into machine aesthetics. Since, we are unable to use visual documentation due to copyright restrictions, here we will mostly reflect on its catalog.

3.2 ToS in juxtaposition to 1934 MoMA Machine Art Exhibition

The 1934 MoMA Machine Art exhibition showcased a series of 'machines, machine parts, scientific instruments, and objects useful in ordinary life' on pedestals as if artworks in five sections as 'household and office equipment, kitchenware, house furnishings and accessories, scientific instruments, and laboratory glass and porcelain' (MoMA, 1934; MoMA, 2016; Wilson et al., 1986; Wilson, 2001). It presented what we could call a *simplicity program* based on five dimensions of beauty in machine art: "geometric, rhythmical, technical and material, visual complexity, and function" (Barr, 1934, no pagination).

Curator Alfred H. Barr (1934) explains these dimensions in the foreword of the exhibition catalog. In geometric beauty, he praises the machine for its ability to create geometric forms to perfection – as in Plato's "always and absolutely" beautiful shapes created by "lathe, ruler, and square" (Plato: Philebus 51 c, cited in MoMA, 1934, no pagination) – drawing a natural bond between the machine and its outcome:

> "Machines are, visually speaking, a practical application of geometry. Forces which act in straight lines are changed in direction and degree by machines which are themselves formed of straight lines and curves. The lever is geometrically a straight line resting on a point. The wheel and axle is composed of concentric circles and radiating straight lines. The watch spring (No. 7) is a spiral. Sphericity and circularity are the geometrical characteristics of a ball bearing (No. 50). Screws, bearing springs (No. 1), and propellers (No. 41) are various —and variously beautiful applications of the helix and helicoid." (Barr, 1934, no pagination)

In rhythmical beauty, Barr (1934) praises static over kinetic rhythm emphasizing how bringing geometrical forms into a certain visual rhythm creates beauty. In technical and material beauty, he talks about sensual (tactile or visual) experiences through appropriate surface treatment of the materials instead of ornamentation. In visual complexity, he mentions how the right amount of complexity keeps interest. In function, he emphasizes the importance of knowing the mechanical or utilitarian function for higher aesthetic pleasure.

The exhibition catalog continues with Philip Johnson's text 'History of Machine Art', where he criticizes craft imitations through machine production as parodies. He states that historically design was done with reference to the machine and technology of the time giving the example of the pottery wheel, however, claims that in the 19th century, this relation was lost, and "technics and design were divorced" (Johnson, 1934, no pagination).

The curatorial texts suggest that the simplicity program of the Machine Art exhibition emerges as a 'descendent' of the machine. Both Barr and Johnson (1934) emphasize the relationship between machines and things, almost like a parent-child notion of shared genetic information, DNA. It is the machines' nature and capabilities that dictate the properties of the simplicity program. Following this principle also in our current context, we would, instead of taking aesthetic principles from elsewhere, consider the intrinsic character of *fluid assemblages* and what kind of aesthetics this implies. For instance, while simpler machines are characterized by their function, and therefore encourage a focus on the efficiency in performing precisely that function, what assemblages 'do' is much more opaque, layered, and relational.

Furthermore, in this exploration, we would look for how *fluid assemblages* differ from things as presented and discussed in the exhibition in their being in terms of form, structure, relation, and purpose. Through these distinctions, we may not only reveal why we need different aesthetic principles today, but also already start to frame a different programmatic space.

As the catalog suggests, the Machine Art exhibition presented straightforward, finished products ready to be used that are produced by a manufacturer using independent machines (MoMA, 1934). On the other hand, *fluid assemblages* are never finished, which creates a distinction around *stability* versus *dynamism*; are always in the making through constant interaction between actors, making every actor in the ecosystem a part of the assemblage, which creates a distinction around *thingness* versus *assemblages*; and are

complex with their visible and hidden layers, which creates a distinction around *perceivable complexity* versus *hidden complexity* (Redström & Wiltse, 2019) (Figure 3).

Another distinction lies in use and use value. The objects presented in the Machine Art exhibition have the same specific functions for all (MoMA, 1934), while *fluid assemblages* become "fundamentally different 'things' framed by different intentions" (Redström & Wiltse, 2019, pg. 155) for different users, referred as *multiintentionality,* and thus offer different values to multiple actors suggesting a distinction around *function* versus *multiintentionality* (Figure 3).

The final distinction is about relating to things. The Machine Art exhibition's catalog (1934) presented the objects with their prices implying that the objects could change ownership for respective prices, while in *fluid assemblages* instead of gaining ownership, ongoing relations are formed through *ToSsphere* agreements (Özçetin & Wiltse, 2023; Redström & Wiltse, 2019) suggesting another distinction around *ownership* versus *relationship* (Figure 3).

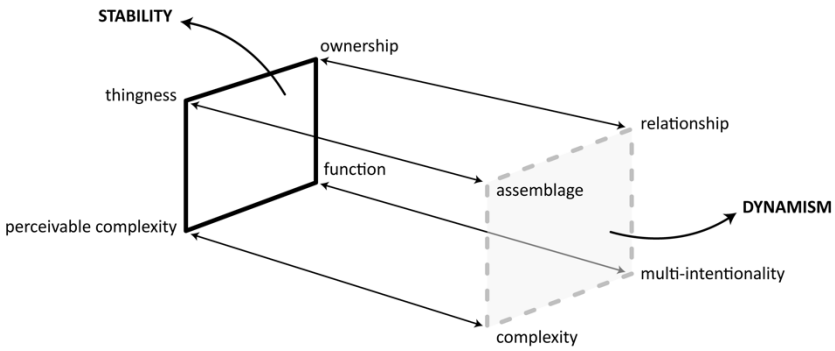


Figure 3 Distinctions drawn between fluid assemblages and things as presented at Machine Art Exhibition

This comparison not only illustrates how contemporary things and machines are different from the past's; but also suggests that their aesthetic principles could be found through this kind of an exploration rather than in the aesthetic principles of the past. The common thread combining the characteristics of *assemblage*, *complexity*, *multiintentionality*, and *relationship* is dynamism as it is inherent to all. This suggests that we are now outside the simplicity program and instead might be looking at programs taking dynamism as the essence (Figure 4).

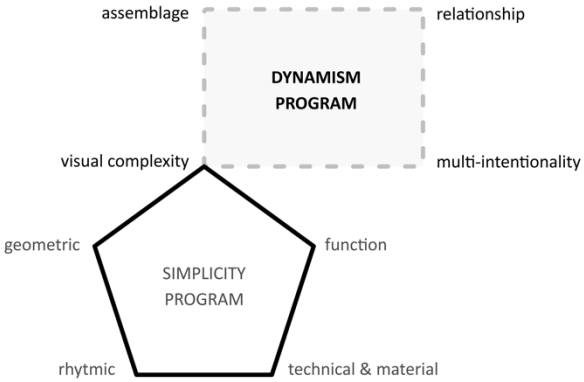


Figure 4 Stepping out of the simplicity program and arriving at the dynamism program

3.3 Reorienting ToS towards Futures

This shift from ownership and distinct relations to specific things, to dynamic relationships governed by *ToSspheres*, raises new kinds of questions for design – and most importantly to what domain or area these questions belong. In the case of the sculptural ornament that was kept from the time when it was a sign of extensive and highly skilled craft, the response had to be in the same sculptural domain. Even if the reason for the ornament was to suggest a higher monetary or social value, and the 'problems' it caused were cluttering and obscuring what characterized the new technologies, the design response was at the level of visual and sculptural form – because that was where it came to expression through design.

The historical success of industrial design may easily lead us to think that this means that most, if not all, design problems need to be solved in the same way: by means of improving visual and sculptural form. We then come to a contemporary issue such as ToS with the idea that we can improve things by making them easier to navigate, visually more appealing, and so on and so forth. The issue of *ToSsphere*, however, is not within the visual or sculptural domain. What we see emerge here instead is the need for an *aesthetics of relating, rather than an aesthetics of use*.

Since much of what we relate to are non-human actors, such as machines and artificial intelligence with intentionalities and agency, embracing more-than-human worldviews by drawing from posthuman theories such as actor-network theory, feminist new materialism, object-oriented ontology seems necessary (Forlano, 2017; Frauenberger, 2019): "Not because humans matter less but because it is no longer exclusively humans that act, design, make use, change, and thus create new possibilities." (Giaccardi & Redström, 2020, p.36). In other words, it might be time to release design from 'user-centered design' and find alternative orientations that is about constructing and maintaining our relationship with digital services.

While much design work needs to be done for an aesthetics of relating program to unfold, *the practice of revealing design* consisting of *relational and power mapping*, *use and users*, and *adjusting appropriateness in case of inappropriateness*, based on posthuman theories as

entanglements, decentering the user, and co-performance, starts to sketch such a design space as it works through the relationality in *fluid assemblages* by making actors, relations, and intentions visible (Özçetin & Wiltse, 2023).

4. Making a method: Programmatic Time Travel

Through this inquiry that tries *[dis/re]orientations* towards pasts, we were able to move away from the limited perspective about ToS that it is a legal or at most a usability problem rather than a design problem. This encouraged us to consolidate our inquiry based on *[dis/re]orientations* into a method to be used in similar inquiries. Reflecting on our actions, we were able to draw out four *[dis/re]orientation* tactics for engaging with an example of a past program to create an alternative programmatic research space.

First, by looking closely to the context at hand, we made an association with the current issue and a past one: today's experience of 'ToS', and 'ornament' in early industrialization. This association was our first disorientation and guided us to look for situated particulars of design artifacts or exhibitions as examples of programs reconfiguring making in industrial production by reflecting on the idea of 'ornament'. As a result, we found ourselves at the 1934 MoMA Machine Art exhibition. We see this as our first tactic:

1. Disorientate – make associations between now and then: Where should one travel to in design histories?

When we arrived at the destination, we examined the materials to understand how they approached the issue at hand. It was how they unpacked the issue we found more valuable than where they arrived as a result. In this case, they highlighted the relationship between things and machines that produce them. This realization guided us to look for the same in the context of ToS, which led us to go to a higher level, fluid assemblages. We see this as our second tactic:

2. Trace the path – focus on the acts, not the outcomes: How did they think in the past to arrive at this outcome?

Then, on one hand looking at *fluid assemblages* and on the other the simplicity program put forward by the exhibition, which extends to this time in certain ways, we decided to look for differences between now and then. This provided us a set of distinctions. We see this as our third tactic:

3. Turn toward past and present (recursively) – define distinctions between now and then: How is the present different from the past?

Finally, from the distinctions we drew out, we looked for the common thread across the distinctions, which was 'dynamism'. This was a reorientation outside the simplicity program, a new starting point for different kinds of research programs to unfold. We see this as our fourth tactic:

4. Reorientate – use connections to create a new programmatic space: What does connect these distinctions?

We, then, suggest that these 4 tactical *[dis/re]orientations* could be framed as a speculative method for making a kind of *Programmatic Time Travel*, which starts from the situated particulars of design and looks for possibilities for connecting design events and interventions that are, seemingly, not that close to each other at all (Figure 5). Indeed, it is the very purpose of this method to make use of such seemingly disconnected times and designs to open up new perspectives. As we turn toward pasts, we avoid design history's big-picture generalizations as major trends and tendencies and the most influential ideas but instead, look for situated particulars of design artifacts or events as examples of programs. It is, thus, about identifying relevant program destinations in design histories to gain insights into present challenges by finding common themes between them.

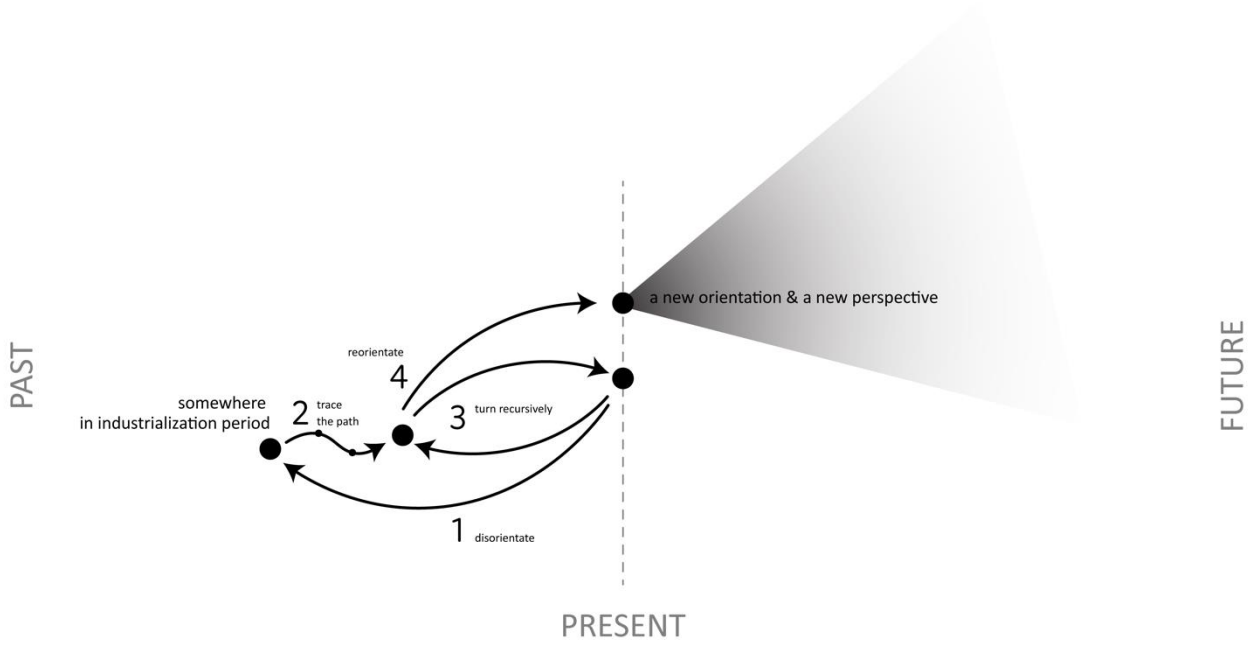


Figure 5. Tactics for Programmatic Time Travel

5. Rethinking design through *Programmatic Time Travels* for alternative futures

We are living in transitional times. The changes we are experiencing are as fundamental as industrialization: "And again, it is how the things are made – the assembling of things – that is transforming the relations between production and consumption, between design and use." (Redström & Wiltse, 2019, p.17). As so-called technological progress fueled by surveillance capitalism relying on generating value about and through use of fluid assemblages (Zuboff, 2019; Özçetin & Wiltse, 2023), dictates what, how, and why and shapes a future often at the expense of humans and non-humans, living and non-living entities, values, dreams, and cultural heritage, it becomes crucial for design to reflect over its ethical foundations and how they manifest, aesthetics. As frictionless interfaces in the

name of achieving user-centered design hide away the complexity in *fluid assemblages*, where multiintentionalities come into play, not only the rift between what things are and how they appear, but also between ethics and aesthetics, grow (Giaccardi & Redström, 2020; Hauser et al, 2021; Redström & Wiltse, 2019).

It might be through looking within design itself, we may start to reconsider design's ethical foundations and transform design to reconnect with it as an epistemological and ontological approach to understanding, knowing, and worlding. Thinking that we may not be able to imagine alternative futures without releasing what no longer serves us effectively, the present superficial understanding and application of user-centered design, we propose turning towards design histories. Through this turn, we may then intervene in how pasts manifests in present and makes possible futures. It is like Dilnot's suggestion to design gestures that retroactively reconfigure the past as affirmative interventions into the paradoxical situations of present:

> "If the past is the transcendental condition for our acts, and the past brought forward into the present is one of the deepest conditioning mechanisms of how we think that we can act, then an act that looks hard at this history and which undoes its false grip on us (which analytically abandons certain unnecessary parameters therefore making space to introduce "novelties" of thought and action) creates an actual new reality by retroactively changing what determines and fixes us and by changing the existing parameters and norms of acting." (Dilnot, 2015, p.212)

These *[dis/re]orientation* tactics towards and through pasts and unfolding juxtapositions that enable a process of defamiliarization opening pathways for alternative ways of thinking, thus could be considered as such an affirmative and retroactive intervention method.

6. Concluding remarks

While most methods for speculations around design concepts and ideas are about futures and projections into emerging possibilities, with the approach introduced here we looked the other way, towards our pasts. The idea was to reimagine designing in a more-than-human world by recovering from the misconceptions we have about user-centered design that we carry along to futures through a reflection in juxtaposition with present. Exploring this approach through the contemporary everyday problem of ToS, we were able to arrive at a new space that we frame as aesthetics of relating.

In this paper we explored ToS by traveling to MoMA's 1934 Machine Art exhibition. This does not however mean that this is the only association one could make. If we look *ToSspheres* from the perspective of power relations, we might then make an association with border control and that would take us to completely different spaces. By turning our expedition into a method based on a series of sequential tactics – *disorientate*, *trace the path*, *turn toward past and present (recursively)*, and *reorientate* – we therefore open it up for engagement with the hope that it could be applied in multiple ways as well as to other contemporary design issues we are dealing with. For example, if we took data as material, could it be possible to explore it in juxtaposition to the ideal 'truth to material', foundational

for Dieter Rams, Bauhaus, and Arts and Crafts movement? What about an association between 'privacy by design' frameworks or Apple's Privacy Nutrition Labels that help users understand data practices and the emergence of design institutions or good design award schemes in the mid-century that aimed to promote design in industry, fight imitation products, and increase export? Other interesting associations could include General Data Protection Regulation (GDPR) article 17 'right to be forgotten' and planned obsolescence or interoperability between different platforms and interchangeability of parts in industrial production. While these associations at first glance could seem irrelevant, as we have tried to exemplify here with the notion of *Programmatic Time Travel*, there are things to be found in creating also seemingly odd connections Recognizing that these contexts and connections however are quite Euro and US-centric, we further encourage extending this approach to other local contexts and start to make global connections.

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About the Authors:

Seda Özçetin is a Marie Skłodowska-Curie PhD fellow at Umeå Institute of Design as part of DCODE Network. Her research focuses on forming and sustaining social contracts as an alternative to Terms of Service.

Johan Redström is Professor in Design at Umeå Institute of Design. He is part of DCODE, and a member of the Committee for Artistic Research at the Swedish Research Council, and the International Advisory board of the Design Research Society.